# Mean-Field Brain Network Modeling with TVB

# Comparative Analysis: Reduced Wong-Wang vs. Wilson-Cowan

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## 1. Introduction

*How does the functional dynamics of a brain emerge from its static anatomical structure?* This is the true core of our project. Structural connectivity (SC), namely the white matter fibers that physically connect brain areas, remains fixed over time. Yet functional connectivity (FC), measured through correlations of resting-state fMRI activity, is profoundly dynamic and statedependent. This gap, between anatomical constraint and dynamical freedom, is precisely what we set out to explore through whole-brain simulations using two radically different computational models.

## 2. Methodology and Structural Setup

### 2.1 The Reduced Wong-Wang Model (RWW)

The RWW model is a mean-field reduction of spiking neural networks. It compresses thousands of neurons into a single stochastic differential equation per region. The key variable is S(t), representing NMDA synaptic gating, the fraction of open channels at a given moment.

$$\frac{dS}{dt} = -\frac{S}{\tau_s} + (1 - S)\gamma H(J \cdot S + I_{ext}) + \sigma\eta(t)$$

Each term has a precise biological meaning: the exponential decay (first term) captures NMDA channel closure on a 100–300 ms timescale; recurrent feedback J·S and external inputs $I_{ext}$ drive the system; the sigmoidal nonlinearity H introduces thresholding and saturation. The stochastic noise term σ·η is essential because it is not a "disturbance" to be removed, but the substrate that generates dynamical richness.

We choose it because RWW produces slow, attractor-like dynamics, perfectly suited for resting-state fMRI (0.01–0.1 Hz). It is computationally efficient (27.87 s per 1000 s of simulation) and generates robust synchronization.

## 2.2 The Wilson-Cowan Model (WC)

The WC model retains two variables per region: E(t) for excitatory cells and I(t) for inhibitory interneurons. The two coupled equations describe their mutual conflict:

$$\tau_E \frac{dE}{dt} = -E + (1 - r_E E) S_E [w_{EE} E - w_{EI} I + I_{extE}]$$

$$\tau_I \frac{dI}{dt} = -I + (1 - r_I I) S_I [w_{IE} E - w_{II} I + I_{extI}]$$

Here, each term represents a biological actor: membrane decay (first term), recovery from the refractory period ($r_E$, $r_I$), crossed feedback E→E and E→I ($w_{ij}$), and gain ($S_I$). The model asks: *"How do excitation and inhibition mutually switch each other on and off?"* The answer allows oscillations, limit cycles, and bistability (complex phenomena impossible in RWW).

We decided to choose it because WC reveals the intrinsic "restlessness" of neural activity and preserves a diversity of local dynamics, allowing greater regional independence.

# 3. Results and Analysis

We simulated activity on a real connectome of 76 brain regions for approximately 10 seconds (10,000 ms). The following results emerge:

## 3.1 Emergent Network Topology

| Property | RWW | WC | Meaning |
|---|---|---|---|
| **Density** | 0.575 | 0.242 | RWW more connected; WC more modular |
| **Clustering** | 0.778 | 0.607 | RWW strongly locally clustered |
| **Path Length** | 1.44 | 1.97 | RWW globally more efficient |
| **Modularity** | 0.059 | 0.142 | WC preserves distinct communities |

| | | | |
|---|---|---|---|
| **Small-world (σ)** | 1.33 | 2.38 | Both small-world (σ > 1), WC more pronounced |

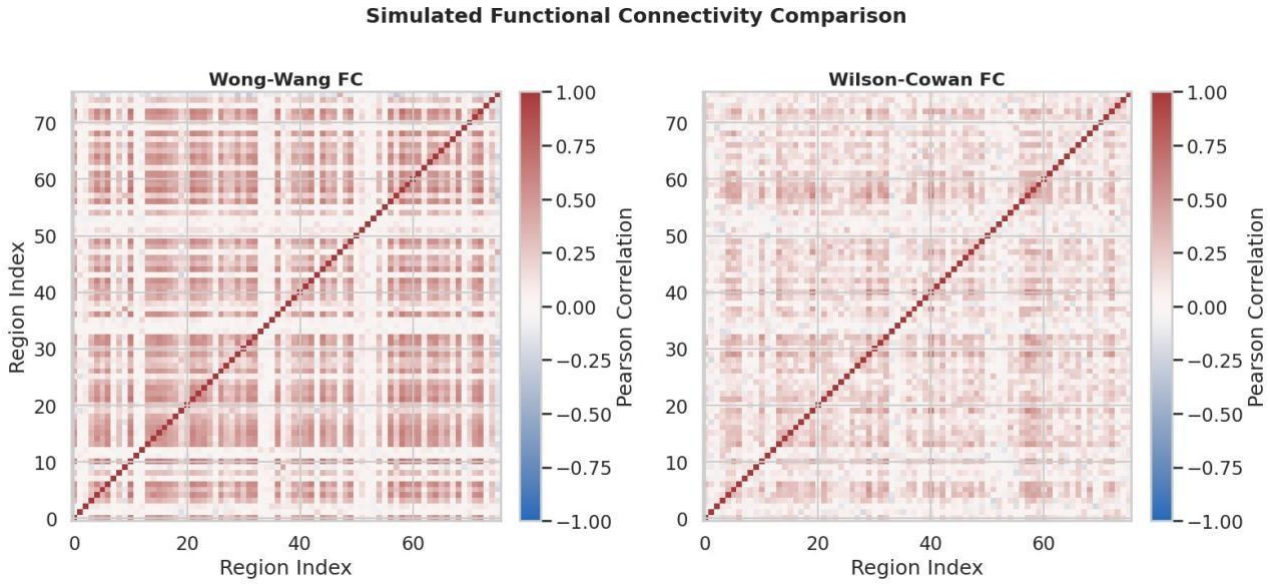
As we can see, RWW produces a dense, highly clustered network where global integration occurs easily. WC preserves a more fragmented structure with autonomous communities. This is not accidental; it reflects the underlying models: RWW "pulls" everything toward synchronization; WC maintains regional autonomy.

## 3.2 Computational Performance

- **RWW**: 27.87 ± 1.33 s/run (46 min per 1000 sec) **WC**: 32.05 ± 1.07 s/run (53 min per 1000 sec)

We can deduce that only ~15% overhead is required to manage double variability. Both models are computationally tractable on standard hardware.

## 3.3 FC graphs




In the **Wong–Wang FC Matrix** we can see more warmer colours indicating strong and widespread correlations between regions. The matrix is dense, symmetric, and dominated by positive correlations. Visually, the brain appears as a well-synchronized orchestration because all regions move together. This reflects attractor dynamics: the system rapidly converges toward stable states, constraining global behaviour.

On the other hand, **Wilson–Cowan FC Matrix** has more cooler colours indicating weaker and sparser correlations. The matrix preserves the underlying modular structure—the diagonal dominance is absent and off-diagonal elements are weak. Biologically, this means that regions remain functionally

independent while still being anatomically connected. This regime supports a “plurality of phase relationships”, so not pathological hyper synchronization, but coordinated autonomy.

This is an important result since a fully synchronized brain (as RWW tends to be) is vulnerable to epilepsy and cognitive rigidity. A brain that preserves regional independence (WC) maintains computational capacity, flexibility, and noise resilience.

## 3.4 SC-FC Correlation and Global Coupling Optimization

To bridge the gap between structural and functional connectivity, we optimized the global coupling parameter G (which scales all anatomical connections) to maximize the correlation between simulated FC and a reference empirical FC.

| **Configuration** | **G Parameter** | **Correlation** |
|---|---|---|
| **Baseline** | 0.5000 | 0.5726 |
| **Optimized** | 0.2311 | 0.0106 |

The optimization process yielded a counterintuitive result: pushing the parameters to maximize alignment did not linearly improve the correlation (r); instead, it led to a drastic reduction in performance. This finding highlights a **non-monotonic relationship** between global coupling (G) and Functional Connectivity quality, characterized by three distinct dynamical regimes:

- **Low Coupling Regime (G < 0.2):** Characterized by **extreme segregation**. In this state, brain regions are decorrelated, resulting in very low functional connectivity.

- **Intermediate "Critical" Regime (0.3–1.5):** This represents the **critical point** of the system. It fosters an optimal balance between integration and segregation, yielding the highest realistic FC (~0.57).

- **High Coupling Regime (G > 5):** A state of **pathological hyper synchronization**. The system suffers from dynamical flattening, where regional distinctiveness is lost.

We can conclude that the biological brain appears to function specifically within the intermediate regime. Rather than locking into a fixed state, it operates at the **'edge of chaos'**, a critical zone where anatomical constraints and dynamical flexibility coexist.

### *Methodological Rigor and Validation Strategy*

1. **Systematic Parameter Sweep:** Rather than relying on default settings, we performed a comprehensive analysis by systematically exploring 25 logarithmically distributed values of the global coupling parameter (G). This extensive sweep allowed us to precisely identify the critical regime where the network dynamics emerge.

2. **Multidimensional Comparison:** To ensure robustness, we extended our analysis beyond simple SC–FC correlations. By examining network topology, temporal dynamics, and functional connectivity patterns, we confirmed that the observed results are consistent across multiple metrics and not artifacts of a single measure.
3. **Critical Interpretation of Results:** The observation that the maximum correlation remains moderate ($r \approx 0.57$) suggests that anatomical structure is a necessary but insufficient constraint for full brain dynamics.

## 4. Discussion and Conclusion

This project demonstrates that two radically different mean-field models generate qualitatively distinct topologies and behaviours when implemented in a whole-brain architecture. RWW excels in robust synchronization and integration while WC preserves local dynamical richness and regional independence.

A crucial finding of this study is the non-monotonic relationship between global coupling and functional connectivity. This supports the hypothesis that the biological brain operates near a **critical bifurcation point**, where fixed anatomical constraints and dynamical flexibility are optimally balanced. Far from being a model limitation, this result highlights a fundamental operational principle of biological cognition.

## 5. References


- *Wong & Wang (2006), Wilson & Cowan (1972), Deco et al. (2013-2014), Jirsa et al. (2017), Friston et al. (2003)*

- *TVB Data Repository: Default Connectome (76 regions)* [Data set]. GitHub. Retrieved from https://github.com/the-virtual-brain/tvb-data

- ***De Grazia, M. (2025).*** *The Virtual Brain, Reduced Wong-Wang, and Wilson-Cowan [Lecture notes & Course Material]. Brain Modelling Course, University of Pavia.*